



HK IMO Training 2021-2022
Problem Set 18
For 19 February 2022



Problem 69. In $\triangle ABC$, its incircle touches the sides BC, CA, AB at D, E, F respectively. Let P be a point on the segment BF such that DF bisects $\angle ADP$. Let Q be a point on the segment CE such that DE bisects $\angle ADQ$. Prove that the circumcircle of $\triangle APQ$ is tangent to the incircle of $\triangle ABC$.

Problem 70. There is an $n \times n$ chessboard where $n \geq 2$. Find the largest positive integer k such that no matter how we colour the cells of the chessboard black or white, the following condition is always met: We can place k non-overlapping dominoes (of size 1×2 or 2×1) on the chessboard such that all of them cover two cells of the same colour, or all of them cover two cells of distinct colours.

Problem 71. Find all positive integers d such that d is the greatest common divisor of

$$\binom{n+1}{n}, \binom{n+2}{n}, \dots, \binom{2n}{n}$$

for some positive integer n .

Problem 72. Find all function(s) $f : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ such that

$$f\left(\frac{f(x)}{f(y)} + 1\right) = f\left(\frac{x}{y} + y + 1\right) - f(y)$$

for any $x, y \in \mathbb{R}^+$.